

STANDARD OPERATING PROCEDURE

BGS Directory of Mines & Quarries

→ **Xtrium Supplier Graph**

SOP Number:	SOP-DS-002
Version:	1.1
Effective Date:	June 4, 2026
Department:	Data Operations
Prepared By:	Xtrium Data Team
Source Dataset:	BGS Directory of Mines and Quarries 2020 (11th Edition)
Source URL:	https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/
Target Schema:	Xtrium Supplier Graph Schema v1.0
Applies To:	External Data Scraping Contractors

Purpose of this Document

This SOP provides complete, step-by-step instructions for extracting operator, mine, and quarry data from the BGS Directory of Mines and Quarries 2020 PDF and transforming each record into a valid Xtrium Supplier Graph Schema JSON object. The objective is to build a material-centric sourcing map — enabling anyone searching for a raw mineral or commodity to locate every UK extraction site and operator associated with it.

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1. Purpose & Scope

This Standard Operating Procedure (SOP) governs the extraction, transformation, and output of data from the British Geological Survey (BGS) Directory of Mines and Quarries (DMQ) 2020 PDF into the Xtrium Supplier Graph Schema JSON format.

The DMQ covers active mineral operations in the United Kingdom during the period 2019–2020. Each site record identifies a physical extraction location, the operator responsible for it, and the commodity produced. The Xtrium goal is material-centric: for any given mineral or commodity, the graph must be able to answer the question — who produces it, where, and under what operator?

1.1 Objectives

- Parse every mine, quarry, pit, oil/gas site, peat working, and mineral handling site from the BGS DMQ 2020 PDF.
- Create one supplier JSON object per unique operating company, not per site.
- Record each site as a `products_offered` entry within its operator's supplier object.
- Map commodity and geological data to product entries so materials can be traced back to source.
- Produce valid JSON conforming to the Xtrium Supplier Graph Schema for direct ingestion.

1.2 Source Dataset Overview

Attribute	Detail
Publication	Directory of Mines and Quarries 2020 — Eleventh Edition
Publisher	British Geological Survey (BGS) / UKRI
Coverage Period	Active mineral operations 2019–2020
Geographic Scope	United Kingdom (England, Scotland, Wales, Northern Ireland, Channel Islands, Isle of Man)
Source Database	BRITPITS — BGS relational database of mineral workings (230,000+ records)
Data Organisation	Firstly by commodity, then by Mineral Planning Authority (MPA), then by operator
Download URL	https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/
Format	Free PDF download (~10.5 MB)
Key Fields per Site	Site name, operator name, operator address/postcode, OS grid reference, MPA, commodity, secondary commodity, end use, geological formation

2. Roles & Responsibilities

Role	Responsibility
PDF Extraction Specialist	Downloads the BGS DMQ 2020 PDF. Parses or manually transcribes site records. Exports structured data (CSV or JSON) for handoff to the Transformation Specialist.
Data Transformation Specialist	Applies this SOP to group site records by operator, build supplier objects, and populate all schema fields. Produces one JSON output file per operator.
QA Reviewer	Runs the validation checklist (Section 11) against each completed JSON file. Rejects non-compliant files and returns them for correction.
Project Lead	Resolves ambiguous operator name conflicts, handles edge cases not covered by this SOP, and accepts final deliverables.

3. Understanding the BGS DMQ Data Structure

Before transforming any data, every team member must understand how the BGS DMQ organises its content, because this directly determines how supplier objects are assembled.

3.1 How the PDF Is Organised

The DMQ PDF is structured in three nested layers:

Layer	Description	Example
1 — Commodity	The outermost grouping. Every entry appears under its commodity heading.	LIMESTONE, SANDSTONE, SILICA SAND, FLUORSPAR, GYPSUM, COAL, etc.
2 — Mineral Planning Authority (MPA)	Within each commodity, sites are listed by the local authority responsible for mineral planning.	Derbyshire County Council, Highland Council, Antrim & Newtownabbey Borough Council
3 — Site Record	The individual mine or quarry entry. Contains site name, operator, address, grid reference, end use, and geology.	Tunstead Quarry, Buxton Lime Industries Ltd, SK089748, Portland Limestone Fm

3.2 Fields Available Per Site Record

Each site entry in the DMQ typically contains some or all of the following fields. Not all fields appear for every site — treat any absent field as null.

DMQ Field	Description	Example Value
Site Name	Name of the mine, quarry, or operation	Tunstead Quarry
Operator Name	Company operating the site	Buxton Lime Industries Ltd
Operator Address	Registered or operational address of the operator	Tunstead, Buxton, Derbyshire, SK17 8TG

DMQ Field	Description	Example Value
OS Grid Reference	Ordnance Survey national grid reference (2-letter + 6-digit format)	SK089748
Mineral Planning Authority	Local authority with planning jurisdiction over the site	Derbyshire County Council
Region Code	BGS region abbreviation covering the site (17 defined regions)	EMD (East Midlands)
Commodity	Primary mineral commodity produced at the site	Limestone
Secondary Commodity	A secondary mineral produced at the same site, if listed	Clay, Chert
End Use	How the extracted commodity is used	Aggregate, Chemical, Agricultural, Cement
Geological Formation	Rock or deposit type being worked	Portland Limestone Formation
Site Type (inferred)	Type of mineral working — infer from site name and DMQ section heading	Quarry, Mine, Pit, Oil Well, Gas Well, Brine Well, Handling Site, Wharf, Processing Plant, Peat Workings

KEY INSIGHT — One Operator, Many Sites

A single operating company (e.g., Tarmac Ltd, CEMEX UK, Aggregate Industries) may operate dozens of sites across different commodities and regions. In the Xtrium schema, the company is ONE supplier object. Each site becomes one entry in that supplier's products_offered array. Always group by operator first, then list sites as products.

3.3 The 17 BGS Regions

Region Code	Region Name	Notes
CHA	Channel Islands	Grid coordinates are UTM
CSCT	Central Scotland	
EEN	East England	
EMD	East Midlands (England)	
IOM	Isle of Man	Supplied with Northern Ireland data
LON	Greater London	
NEA	North-east England	
NIR	Northern Ireland	
NSCT	North Scotland	
NWE	North-west England	

Region Code	Region Name	Notes
NWLS	North Wales	
SEA	South-east England	
SSCT	South Scotland	
SWE	South-west England	CHA data included here
SWLS	South Wales	
WMD	West Midlands (England)	
YHU	Yorkshire and Humberside (England)	

4. Schema Mapping Strategy

This section explains the fundamental design decisions for how BGS data maps into the Xtrium Supplier Graph Schema. Read this before attempting any data entry.

4.1 Core Design Principle — Operator as Supplier

In the Xtrium schema, the supplier object represents a company, not a physical site. The BGS DMQ operator is the company. The operator's individual mine or quarry sites map to `products_offered` entries within that company's supplier object.

BGS DMQ Concept	Xtrium Schema Destination	Notes
Operator (company)	<code>suppliers[]</code> — one object per unique operator	Primary mapping unit
Operator address / postcode	<code>headquarters_location</code>	Use as proxy for HQ if no better data
Individual site (mine/quarry)	<code>products_offered[]</code> — one entry per site per commodity	Each site is a 'product' of the supplier
Commodity produced at site	<code>products_offered[].product_name</code> AND <code>products_offered[].category</code>	The mineral is the product
Secondary commodity at site	Additional <code>products_offered[]</code> entries — one per secondary commodity	Same site entry; separate product entries. See Rule 2.
End use of commodity	<code>products_offered[].grade</code>	End use functions as a grade/variant designation
Geological formation	<code>products_offered[].product_id</code> (as site code) or raw	No direct schema field; treat as supplemental product data
OS Grid Reference	<code>manufacturing_sites[].location</code>	Primary location identifier for the site
Mineral Planning Authority	<code>manufacturing_sites[].raw</code>	Regulatory context, not a standalone schema field

BGS DMQ Concept	Xtrium Schema Destination	Notes
BGS Region Code	manufacturing_sites[].country (inferred) + raw	Infer country from region; record raw code
Site type (mine vs quarry etc.)	manufacturing_sites[].site_type	See full permitted list in Section 5.2

4.2 Supply Chain Tier Assignment

Operators in the BGS DMQ are active extractors — they are raw material producers. Apply the following tier logic:

Operator Type	supply_cha in_tier	Rationale
Mine / quarry / pit operator (direct extraction)	1	Tier 1 = raw material producer. All BGS DMQ operators extracting at source.
Peat workings (direct extraction)	1	Peat extraction is primary production. Same tier as quarry operators.
Mineral handling site only (no extraction)	2	Tier 2 = processor/redistributor. Handling sites process or redistribute rather than extract.
Processing plant (lime works, tileworks, salt plant)	2	Tier 2 = processor. These sites transform extracted material, not extract at source.
Spoil heap / bing operations	2	Tier 2 = secondary processor. Bings rework existing waste material, not primary extraction.
Recycling facility (aggregate recycling, waste mgmt.)	2	Tier 2 = processor. Recycled aggregates are processed from waste, not extracted from ground.
Wharf / maritime handling site	2	Tier 2 = redistributor. Wharves handle imported or transferred material.
Distributor / reseller (if identifiable)	3	Tier 3 = distributor. Rare in DMQ but apply if operator description indicates distribution only.

⚠ IMPORTANT Default to tier 1 for all standard quarry, mine, and pit operators unless the site record specifically describes the operation as handling, processing, recycling, or distribution only.

4.3 Industry Sector Mapping [AMENDED v1.1]

This table has been extended in v1.1 to include Clay and Shale, Basalt (NI), Igneous & Metamorphic Rock, Recycled aggregates, and Peat — five commodity types not addressed in the original SOP.

BGS Commodity Category	Xtrium industry_sector
Construction aggregates (sand, gravel, crushed rock, limestone for aggregate)	construction minerals
Industrial minerals (silica sand, ball clay, china clay, fluorspar, gypsum, potash, salt)	industrial minerals
Clay and Shale (brick, tile, ceramics)	industrial minerals
Building stone (sandstone, granite, slate, marble quarried for building)	building stone
Cement and lime raw materials (limestone/chalk for cement, quicklime)	cement and lime
Coal (surface and underground)	coal
Oil and gas extraction (onshore)	oil and gas
Metals (iron ore, tin, copper, lead-zinc, gold)	metals mining
Basalt (NI) — used primarily as aggregate in Northern Ireland	construction minerals
Igneous & Metamorphic Rock — aggregate and building stone	construction minerals
Recycled aggregates (crushed, reprocessed from waste or demolition material)	recycled aggregates
Peat (extraction from peat workings)	peat
Mineral handling sites (wharves, import terminals)	mineral handling

5. Field-by-Field Mapping Reference

5.1 Supplier-Level Fields

Schema Field	BGS Source	Transformation Rule
supplier_id	None	Always null — assigned by load_to_neo4j.py
duns_number	None	null — not in BGS DMQ; may be resolvable post-extraction via D&B lookup
company_name	Operator Name	Copy exactly as printed in DMQ. Do not abbreviate or expand.
canonical_name	Derived	Lowercase, spaces to hyphens, & to 'and', remove remaining punctuation. See Section 7.2 for full rules.
headquarters_location	Operator Address	Extract town/city and country. Format: 'City, Country'. e.g. 'Buxton, United Kingdom'

Schema Field	BGS Source	Transformation Rule
website	None	null — not in DMQ. May be populated post-extraction via web search.
company_description	None / inferred	null from DMQ alone. Optionally populate with brief operator summary from public sources.
industry_sector	Commodity (inferred)	Apply sector mapping table in Section 4.3. Use primary commodity type.
supply_chain_tier	Site type (inferred)	Apply tier rules in Section 4.2. Default: 1 for extraction sites.
typical_lead_time_days	None	Always null — not stated in DMQ.
is_verified	N/A	Always false from scraper.
certifications	None	Empty array [] — not in DMQ.
regulation_references	None	Empty array [] — not in DMQ.
certifications_raw	None	null — not in DMQ.

5.2 Manufacturing Sites Array [AMENDED v1.1]

Each physical mine or quarry operated by the company becomes one entry in the manufacturing_sites array. Use the OS Grid Reference as the primary location identifier.

Schema Field	BGS Source	Transformation Rule
location	Site Name + OS Grid Reference	Format: '{Site Name} ({OS Grid Ref})'. e.g. 'Tunstead Quarry (SK089748)'
country	BGS Region Code (inferred)	CSCT/NSCT/SSCT → 'Scotland'; NIR → 'Northern Ireland'; NWLS/SWLS → 'Wales'; CHA → 'Channel Islands'; IOM → 'Isle of Man'; all others → 'England'
site_type	Site type label (inferred)	Infer from site name and DMQ section heading. See Section 10 for edge case guidance. Permitted values (v1.1 extended list): quarry mine pit oil well gas well brine well handling site wharf processing plant spoil heap recycling facility peat workings
raw	Full verbatim DMQ entry	Copy the complete text of the site record as it appears in the PDF, including MPA, end use, and formation.

5.3 Products Offered Array [AMENDED v1.1]

Primary commodity: Each unique combination of site + commodity becomes one products_offered entry.

Secondary commodity: If the DMQ lists a secondary commodity for a site, create an additional products_offered entry for it, using the secondary commodity as product_name. Use a different abbreviation suffix in product_id to ensure uniqueness.

If a single site produces two distinct end uses for the same commodity (e.g., limestone for aggregate AND limestone for cement), create two separate products_offered entries with different grade values.

Schema Field	BGS Source	Transformation Rule
product_name	Commodity (primary or secondary)	Commodity name in Title Case. e.g. 'Limestone', 'Silica Sand', 'Coal (Surface Mined)'
grade	End Use	Map end use as grade/variant. e.g. 'Aggregate', 'Chemical', 'Agricultural', 'Cement raw material', 'Graded'. Graded variants (e.g., 'Graded sand & gravel') use grade = 'Graded' — do not create a separate product_name.
product_id	OS Grid Ref + Commodity code	Format: '{GridRef}_{CommodityAbbrev}'. e.g. 'SK089748_LS'. Secondary commodities use a different abbreviation suffix to maintain uniqueness.
category	BGS Commodity Section heading	Copy the section heading. e.g. 'LIMESTONE', 'SILICA SAND', 'FLUORSPAR'
source_url	BGS DMQ download page	Always: 'https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/'
datasheet_url	None	null — PDFs are publication-level, not product-level
cross_graph_material_id	None	Always null during extraction — populated post-load by the ID resolver

5.4 Sources Array

Every supplier object must include at least one source citation pointing to the BGS DMQ publication. This is a government/official source and is always tier1.

```
"sources": [
  {
    "source_name": "BGS Directory of Mines and Quarries 2020, 11th Edition",
    "source_url": "https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/",
    "doi": null,
    "tier": "tier1"
  }
]
```

5.5 Data Completeness Flags

Set these flags as shown below for all BGS DMQ records. These values are fixed — they signal downstream agents what can and cannot be sourced from this publication.

Flag Field	Value	Reason
review_score	manual_only	Supplier performance reviews are not in DMQ; require human research
defect_rate_ppm	manual_only	Not available from any public source; requires direct supplier contact
on_time_delivery_rate	manual_only	Not available from any public source; requires direct supplier contact
pricing	api_only	Commodity pricing available via market price APIs, not on operator websites
inventory_levels	api_only	Mine output statistics available from BGS World Mineral Statistics API

6. Non-Negotiable Operating Rules

RULE 1 — ONE SUPPLIER OBJECT PER OPERATOR

Group all sites belonging to the same operating company into a single supplier object. Never create two supplier objects for the same legal entity. If an operator appears under multiple commodities or regions, consolidate all their sites into the one object.

RULE 2 [AMENDED v1.1] — ONE products_offered ENTRY PER SITE PER COMMODITY PER END USE

A site producing limestone for aggregate AND cement creates two products_offered entries with different grade values. A site that also lists a secondary commodity (e.g., clay alongside limestone) creates additional products_offered entries — one per secondary commodity. A site producing only one commodity with one end use creates one entry. Never collapse multiple sites into one products_offered entry.

RULE 3 — EXACT TEXT PRESERVATION IN raw

The raw field on every manufacturing_sites entry must contain the verbatim text of the DMQ site record as it appears in the PDF. Do not paraphrase, summarise, or clean this field. It is the audit trail back to the source.

RULE 4 — NULL FOR ABSENT DATA — NEVER INVENT

If the DMQ does not state a value for a field (e.g., website, lead time, certifications), set that field to null or []. Do not search the web to fill in gaps during extraction — that is a separate enrichment step.

RULE 5 — source_url IS ALWAYS THE PUBLICATION PAGE

The `source_url` for every `products_offered` entry and every `sources[]` entry must point to the BGS DMQ 2020 download page — not to a search result, not to the BGS homepage, not to an operator's website. This is the authoritative citation.

RULE 6 — `is_verified` IS ALWAYS false

The `is_verified` flag is set to false by all scrapers without exception. It is set to true only after a human reviewer has independently verified the operator information. Never set this to true during extraction.

7. Operator Deduplication Rules

The DMQ lists operators repeatedly across different commodities and regions. Before creating a new supplier object, always check whether the operator already exists in your working dataset.

7.1 Matching Rules (apply in order)

- Exact `company_name` match → same supplier object. Add new sites to existing `products_offered` and `manufacturing_sites` arrays.
- Near match (e.g., 'Tarmac Trading Ltd' vs 'Tarmac Ltd') → flag for Project Lead review. Do not auto-merge.
- Same address + similar name → likely same entity. Flag for Project Lead review.
- Same name, different postcode region → may be a subsidiary or regional division. Flag for Project Lead review.

⚠ IMPORTANT Large aggregate producers (Tarmac, CEMEX, Aggregate Industries, Hanson, Breedon Group) operate across dozens of sites. You will encounter these names repeatedly as you work through different commodity sections. Always add new sites to the existing supplier object. Do not create duplicate entries.

7.2 `canonical_name` Transformation [AMENDED v1.1]

Apply all transformations below in order. The key v1.1 addition is the explicit conversion of & to 'and' before other processing.

- Convert full `company_name` to lowercase.
- **Convert & to 'and'** before any other processing. e.g. 'A & D Sutherland' — the & becomes 'and', not silently dropped. Silently dropping & risks false matches with unrelated entities sharing the same initials.
- Replace spaces with hyphens.
- Remove all remaining punctuation (commas, dots, parentheses, apostrophes).
- Keep common suffixes as-is: ltd, plc, llp, uk, group.

company_name	canonical_name
Tarmac Ltd	tarmac-ltd
CEMEX UK Operations Ltd	cemex-uk-operations-ltd
Aggregate Industries UK Ltd	aggregate-industries-uk-ltd
Hanson Quarry Products Europe Ltd	hanson-quarry-products-europe-ltd
Breedon Group plc	breedon-group-plc
Sibelco UK Ltd	sibelco-uk-ltd
Imerys Minerals Ltd	imerys-minerals-ltd
British Gypsum Ltd	british-gypsum-ltd
A & D Sutherland	a-and-d-sutherland

8. Complete Worked Examples

8.1 Single-Site Operator — Specialist Industrial Mineral

Source DMQ Entry (verbatim from PDF)

```
FLUORSPAR
Derbyshire County Council
  Glebe Mine
  Cavendish Mill Fluorite Ltd
  Cavendish Mill, Stoney Middleton, Hope Valley, Derbyshire, S32 4TF
  Grid Ref: SK222756
  End Use: Acid grade fluorspar, Ceramic grade fluorspar
  Formation: Blue John Cavern Formation (Derbyshire limestone)
```

Output JSON

```
{
  "supplier_id": null,
  "duns_number": null,
  "company_name": "Cavendish Mill Fluorite Ltd",
  "canonical_name": "cavendish-mill-fluorite-ltd",
  "headquarters_location": "Hope Valley, United Kingdom",
  "website": null,
  "company_description": null,
  "industry_sector": "industrial minerals",
  "supply_chain_tier": 1,
  "typical_lead_time_days": null,
  "manufacturing_sites": [
    {
      "location": "Glebe Mine (SK222756)",
      "country": "England",
      "site_type": "mine",
    }
  ]
}
```

```

    "raw": "Glebe Mine, Cavendish Mill Fluorite Ltd, Stoney Middleton,
Derbyshire. Grid: SK222756. MPA: Derbyshire County Council. End Use: Acid grade
fluorspar, Ceramic grade fluorspar. Formation: Blue John Cavern Formation."
  },
  "certification_references": [],
  "certifications_raw": null,
  "regulation_references": [],
  "products_offered": [
    {
      "product_name": "Fluorspar",
      "grade": "Acid grade",
      "product_id": "SK222756_FLU_AG",
      "category": "FLUORSPAR",
      "source_url":
"https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/",
      "datasheet_url": null,
      "cross_graph_material_id": null
    },
    {
      "product_name": "Fluorspar",
      "grade": "Ceramic grade",
      "product_id": "SK222756_FLU_CER",
      "category": "FLUORSPAR",
      "source_url":
"https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/",
      "datasheet_url": null,
      "cross_graph_material_id": null
    }
  ],
  "is_verified": false,
  "data_completeness_flags": {
    "review_score": "manual_only", "defect_rate_ppm": "manual_only",
    "on_time_delivery_rate": "manual_only", "pricing": "api_only",
    "inventory_levels": "api_only"
  },
  "sources": [{ "source_name": "BGS Directory of Mines and Quarries 2020, 11th
Edition",
    "source_url":
"https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/",
    "doi": null, "tier": "tier1" }]
}

```

8.2 Multi-Site, Multi-Commodity Operator — Large Aggregate Producer

Source DMQ Entries — Same Operator Across Two Commodity Sections

```

LIMESTONE (CRUSHED ROCK)
Derbyshire County Council
Tunstead Quarry
Tarmac Ltd
Tunstead, Buxton, Derbyshire, SK17 8TG
Grid Ref: SK089748 | End Use: Aggregate, Cement | Formation: Eyam Limestone
Fm

GRITSTONE / SANDSTONE
West Yorkshire (Kirklees MBC)
Crosland Moor Quarry
Tarmac Ltd

```

Huddersfield, West Yorkshire, HD4 7AE
Grid Ref: SE107138 | End Use: Aggregate | Formation: Rough Rock

Output JSON — Single Supplier Object, Three products_offered Entries

```
{
  "company_name": "Tarmac Ltd",
  "canonical_name": "tarmac-ltd",
  "headquarters_location": "Buxton, United Kingdom",
  "industry_sector": "construction minerals",
  "supply_chain_tier": 1,
  "manufacturing_sites": [
    { "location": "Tunstead Quarry (SK089748)", "country": "England", "site_type":
      "quarry",
      "raw": "Tunstead Quarry, Tarmac Ltd, Tunstead, Buxton, SK17 8TG. MPA:
      Derbyshire CC. End Use: Aggregate, Cement. Formation: Eyam Limestone Fm." },
    { "location": "Crosland Moor Quarry (SE107138)", "country": "England",
      "site_type": "quarry",
      "raw": "Crosland Moor Quarry, Tarmac Ltd, Huddersfield, HD4 7AE. MPA:
      Kirklees MBC. End Use: Aggregate. Formation: Rough Rock." }
  ],
  "products_offered": [
    { "product_name": "Limestone", "grade": "Aggregate",
      "product_id": "SK089748_LS_AGG", "category": "LIMESTONE",
      "source_url":
      "https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/"
    },
    { "product_name": "Limestone", "grade": "Cement raw material",
      "product_id": "SK089748_LS_CEM", "category": "LIMESTONE",
      "source_url":
      "https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/"
    },
    { "product_name": "Sandstone", "grade": "Aggregate",
      "product_id": "SE107138_SS_AGG", "category": "GRITSTONE / SANDSTONE",
      "source_url":
      "https://www.bgs.ac.uk/mineralsuk/download/directory-of-mines-and-quarries-2020/"
    }
  ],
  "is_verified": false,
  ...
}
```

9. Step-by-Step Transformation Workflow

Follow these steps in sequence. Complete Step 1 before starting Step 2. Do not attempt to transform and extract simultaneously.

Step	Name	Action
1	Download & Prepare PDF	Download the BGS DMQ 2020 PDF from the BGS MineralsUK download page. Confirm the file is the 2020 Eleventh Edition (file size ~10.5 MB). Save locally with filename: BGS_DMQ_2020_11thEd.pdf.
2	Parse PDF Structure	Open the PDF and identify all commodity section headings (all-caps, e.g., LIMESTONE, SILICA SAND, GYPSUM). For each section, note

Step	Name	Action
		the sub-groupings by Mineral Planning Authority. Use a PDF extraction tool (e.g., pdfplumber, Camelot, or Adobe Acrobat Export) to extract text by section. Validate that all 17 regions are represented.
3	Build Operator Index	Scan the entire PDF and compile a deduplicated list of all unique operator names before transforming any records. This index prevents duplicate supplier objects. Record: operator name, first site encountered, regions present.
4	Process Commodity by Commodity	Work through each commodity section from beginning to end. Do not skip sections. For each site record: (a) Identify the operator; (b) Look up operator in your index; (c) If new — create a new supplier object; (d) If existing — retrieve the existing object; (e) Add site to manufacturing_sites; (f) Add one products_offered entry per commodity per end use. If a secondary commodity is listed, add an additional products_offered entry for it.
5	Populate Supplier-Level Fields	After all sites for an operator are added, populate: headquarters_location (from most common address), industry_sector (from primary commodity per Section 4.3), supply_chain_tier (per Section 4.2), canonical_name (per Section 7.2).
6	Set Fixed Fields	For every supplier object, set: is_verified = false; all data_completeness_flags per Section 5.5 defaults; sources array with BGS DMQ tier1 citation.
7	Handle Handling Sites	Separately process any mineral handling sites, wharves, processing plants, and recycling facilities. Assign supply_chain_tier = 2. Their products_offered entries should reflect the minerals they handle, with grade = 'Import/handling' where end use is not specified.
8	Validate Each Object	Run the validation checklist in Section 11 against each completed supplier object before saving. Do not proceed to the next operator until the current one passes.
9	Assemble Output	Wrap all supplier objects in the top-level suppliers array: { "suppliers": [...all objects...] }. Write to file: bgs_dmqa_2020_suppliers.json.
10	Final Batch QA	Confirm total supplier object count is plausible given the PDF size. Cross-check 10 randomly selected operators to confirm all their sites appear across commodity sections. Submit to QA Reviewer.

10. Edge Cases & Special Handling [AMENDED v1.1]

This section has been extended in v1.1 to cover site_type edge cases, recycling facilities, peat workings, secondary commodities, and graded variants raised by contractor questions.

Situation	Required Action
Operator listed with no address	Set headquarters_location to null. Use region code to infer country for manufacturing_sites.country.
Site with multiple end uses listed (e.g., 'Aggregate, Chemical, Filler')	Create one products_offered entry per end use — all pointing to the same site (same product_id base, different grade suffix).
Operator name appears with slight variation across sections (e.g., 'Tarmac Ltd' vs 'Tarmac Trading Ltd')	Do not auto-merge. Flag to Project Lead. Keep separate until confirmed.
Site name is generic or missing (e.g., unnamed quarry, blank field)	Use grid reference as site name: 'Unnamed Site ({Grid Ref})'. Never leave location blank.
Oil and gas wells — no fixed site name structure	Use well identifier or licence block as location. site_type = 'oil well' or 'gas well'. Commodity = 'Natural Gas' or 'Crude Oil'.
Brine wells and solution mining	site_type = 'brine well'. Commodity = 'Salt' or 'Brine'. supply_chain_tier = 1.
Same physical site, different operator sections (site changed hands)	Create separate products_offered entries under each operator. Note the discrepancy in raw.
Operator has sites in both England and Scotland	Two manufacturing_sites entries with different country values. Both within the same supplier object.
Channel Islands (CHA) sites	country = 'Channel Islands'. Note in raw that grid coordinates are UTM, not OS National Grid.
Handling site with no extraction, commodity listed as 'Various'	product_name = 'Various Minerals'. grade = 'Import'. supply_chain_tier = 2.
Site name contains 'Pit' (e.g., Marston's Pit, sand pit, clay pit)	site_type = 'pit'. Used for surface workings of sand, gravel, and clay that are distinct from hard-rock quarries. supply_chain_tier = 1. Commodity and grade follow the standard rules.
Site name contains 'Wharf' or is a maritime/river handling point (e.g., Derryclone Wharf)	site_type = 'wharf'. supply_chain_tier = 2. product_name reflects the mineral handled. grade = 'Import' if no end use is stated. industry_sector = 'mineral handling'.
Site is a processing or manufacturing facility (e.g., Babylon Tileworks, Crundale Lime Works, Porthkerris Sea Salt Plant)	site_type = 'processing plant'. supply_chain_tier = 2. These facilities transform an extracted mineral into a processed product; they do not extract at source. Apply the correct site_type per site even if the same operator also operates quarries — each site entry is independent.
Site is a spoil heap or bing (e.g., Drumshoreland Bing — coal waste tip reworked for aggregate or fill)	site_type = 'spoil heap'. supply_chain_tier = 2. product_name = 'Colliery Spoil' or the specific material. grade = 'Fill' or 'Aggregate'. industry_sector = 'coal'.
Recycling or waste management facility (e.g., Myers Road Waste Recycling Facility, Southwood Waste Management Facility)	site_type = 'recycling facility'. supply_chain_tier = 2. industry_sector = 'recycled aggregates' if the output is recycled aggregate material. product_name = 'Recycled Aggregate' or the specific material handled.

Situation	Required Action
Peat workings (e.g., Lindow Moss Peat Workings)	site_type = 'peat workings'. supply_chain_tier = 1. industry_sector = 'peat'. product_name = 'Peat'. grade reflects end use (e.g., 'Horticultural', 'Fuel').
End use is described as 'Graded [commodity]' (e.g., 'Graded sand & gravel')	Do NOT create a separate product_name for the graded variant. Keep product_name = the base commodity (e.g., 'Sand and Gravel'). Set grade = 'Graded' or the full end-use phrase. Grading is a processing variant, not a distinct commodity.
Secondary commodity listed at a site (e.g., Clay listed alongside Limestone)	Create an additional products_offered entry with product_name = the secondary commodity. Use the secondary commodity abbreviation in the product_id suffix to ensure uniqueness (e.g., 'SK089748_CL' for clay at that grid ref). The manufacturing_sites entry remains the same — do not duplicate it.

11. Pre-Submission Validation Checklist

Run this checklist against every supplier object before submitting the batch. Reject any object that fails one or more items.

⚠ IMPORTANT Work through each item in order for every supplier object. A single failed item means the object must be corrected before submission. Do not submit incomplete batches — complete all objects or none.

Required Fields

- company_name is present and non-empty.
- products_offered array contains at least one entry.
- sources array contains at least one tier1 BGS DMQ entry.
- Each products_offered entry has product_name and source_url populated.
- Each manufacturing_sites entry has location and raw populated.

Identity & Classification

- canonical_name is lowercase-hyphenated, derived from company_name (& converted to 'and' per Section 7.2).
- supply_chain_tier is set (1 for extraction/peat; 2 for handling/processing/recycling/wharf/spoil heap).
- industry_sector is set and matches primary commodity per Section 4.3 (including new sectors: recycled aggregates, peat).
- headquarters_location is in 'City, Country' format or null if not available.

Site-Level Accuracy

- Each manufacturing_sites entry has site_type set to a permitted value from the list in Section 5.2.
- Each manufacturing_sites entry has country set or inferred from region code.
- raw field on every manufacturing_sites entry contains full verbatim DMQ text.
- OS Grid Reference appears in either location string or raw field.

Product-Level Accuracy

- product_id is unique per site-commodity-grade combination.
- Secondary commodities have their own products_offered entry with a distinct product_id suffix.
- Graded variants (e.g., 'Graded sand & gravel') use grade = 'Graded' — not a separate product_name.
- grade reflects the DMQ end use value for that entry.
- cross_graph_material_id is null (not populated during extraction).
- source_url is exactly the BGS DMQ 2020 download page URL.

Fixed / System Fields

- is_verified is false.
- supplier_id is null.
- duns_number is null (DMQ does not provide this).
- website is null (unless separately researched — flag if so).
- All data_completeness_flags match defaults in Section 5.5.

No Duplicates

- No two supplier objects share the same company_name or canonical_name.
- All sites for the same operator are in one object, not spread across multiple.

12. Common BGS Commodity Reference [AMENDED v1.1]

This table has been extended in v1.1 to include Clay and Shale, Basalt (NI), Igneous & Metamorphic Rock, Recycled aggregates, and Peat (shown at bottom of table).

Commodity (as in DMQ)	product_name	industry_sector	Typical End Uses (→ grade)
Limestone (crushed rock)	Limestone	construction minerals	Aggregate, Cement, Agricultural, Chemical, Filler
Sandstone / Gritstone	Sandstone	construction minerals / building stone	Aggregate, Building Stone

Commodity (as in DMQ)	product_name	industry_sector	Typical End Uses (→ grade)
Sand and Gravel	Sand and Gravel	construction minerals	Concrete Aggregate, Asphalt, Fill, Graded
Chalk	Chalk	construction minerals / cement and lime	Agricultural Lime, Cement raw material
Granite	Granite	construction minerals / building stone	Aggregate, Building Stone
Slate	Slate	building stone	Roofing Slate, Aggregate
Silica Sand	Silica Sand	industrial minerals	Glass, Foundry Sand, Frac Sand
Ball Clay	Ball Clay	industrial minerals	Ceramics, Sanitaryware
China Clay (Kaolin)	China Clay	industrial minerals	Paper coating, Ceramics, Filler
Fluorspar	Fluorspar	industrial minerals	Acid grade, Ceramic grade, Metallurgical grade
Gypsum / Anhydrite	Gypsum	industrial minerals	Plasterboard, Cement retarder, Agricultural
Potash	Potash	industrial minerals	Fertiliser
Salt (Rock Salt / Brine)	Salt	industrial minerals	De-icing, Food grade, Chemical
Coal (Surface)	Coal	coal	Power Generation, Industrial
Coal (Underground)	Coal	coal	Power Generation, Industrial, Coking
Natural Gas (Onshore)	Natural Gas	oil and gas	Energy
Crude Oil (Onshore)	Crude Oil	oil and gas	Refinery feedstock
Iron Ore	Iron Ore	metals mining	Steel making
Barytes	Barytes (Barite)	industrial minerals	Drilling fluid, Chemical, Filler
Clay and Shale	Clay / Shale	industrial minerals	Brickmaking, Tile manufacture, Ceramics
Basalt (NI)	Basalt	construction minerals	Aggregate
Igneous & Metamorphic Rock	Igneous Rock / Metamorphic Rock	construction minerals	Aggregate, Building Stone
Recycled Aggregates	Recycled Aggregate	recycled aggregates	Aggregate, Fill, Sub-base
Peat	Peat	peat	Horticultural, Fuel

13. Document Control

Version	Date	Author	Changes
1.0	April 2, 2026	Xtrium Data Team	Initial release
1.1	June 4, 2026	Xtrium Data Team	Contractor clarifications incorporated (all rulings confirmed by Project Lead): • Q1 Secondary commodity: treated as separate products_offered entry — same site, new product_name (Sections 4.1, 5.3, Rule 2) • Q2 Recycling facilities: site_type = 'recycling facility', supply_chain_tier = 2, industry_sector = 'recycled aggregates' (Sections 4.2, 5.2, 10) • Q3 Ampersand in operator names: convert & to 'and' in canonical_name before all other processing (Section 7.2) • Q4 Industry sector added for: Clay/Shale (industrial minerals), Basalt NI (construction minerals), Igneous & Metamorphic Rock (construction minerals), Recycled aggregates (recycled aggregates), Peat (peat) — (Sections 4.3, 12) • Q5 Graded sand & gravel: grade only, not a separate product_name (Sections 5.3, 10) • Q6 Extended site_type permitted list: added pit, wharf, processing plant, spoil heap, recycling facility, peat workings (Sections 5.2, 10) • Q7 Peat workings: industry_sector = 'peat', site_type = 'peat workings', supply_chain_tier = 1 (Sections 4.2, 4.3, 10, 12)

For corrections or unresolved mapping questions, contact the Xtrium Data Team at raghu@xtrium.ai. All amendments require Project Lead approval before taking effect.

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